# Sanzhar Sovet

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## Summary

AI/ML Engineer with B.Sc. and M.Sc. degrees in Data Science and **3+ years of experience** in **healthcare data analytics** and **ML engineering** across both academic and industry environments. Experience covers applied ML research, LLM fine-tuning, MLOps infrastructure for production, and AI agent development for business needs.

## Education

**Astana IT University** — Astana, Kazakhstan
*Master's degree in Applied Data Analytics — Full Grant, GPA 3.6* — *2023 – 2025*

**Astana IT University** — Astana, Kazakhstan
*Bachelor's degree in Big Data Analysis — Full Grant, GPA 3.3* — *2020 – 2023*

## Experience

**AI/ML Engineer | MLOps** — Oct 2025 – Present
*Global Optima Technologies / Atlanti AI (founded by an ex-Meta/Amazon senior ML engineer)*

- Deployed and maintained **multimodal AI models** in production, covering **API development**, **inference optimization**, **load testing**, and **logging and resource monitoring** (GPU, CPU, RAM)
- Built **AI agent platforms** end to end, covering system architecture, **multi-agent orchestration**, **custom agentic workflows**, LLM deployment, external integrations, and agent evaluation

**Junior Researcher** — Mar 2025 – Present
*AI Research Center at Astana IT University*

- Developed **ML models** for EEG-based attention monitoring as part of NeuroPro.kz
- Authored publications on EEG-based cognitive states and ML methods in **Q1/Q2 international journals**

**Healthcare Data Analyst** — Mar 2024 – Aug 2025
*National Research Center for Healthcare Development*

- Managed and maintained national healthcare databases on **MS SQL Server**
- Developed **Power BI dashboards** (view report) used by **Ministers / Vice-Ministers of Healthcare** of Kazakhstan
- Increased department efficiency by developing a **web scraping** and **Excel automation program** (view project), which saved **2–3 workdays per month** for each of **8 employees**

**Healthcare Data Analyst** — Aug 2023 – Mar 2024
*Ryte.ai*

- Designed database schemas and integrated external healthcare sources for two healthcare projects to ensure **ML-ready datasets**, with pipelines documented in **Azure**
- Prepared **well-structured datasets** in collaboration with Data Engineers, AI specialists, Data Curation, and QA teams
- Analyzed large-scale healthcare datasets in **Databricks** (Python, SQL, PySpark), delivering accurate statistics and reports that enabled data-driven decision-making across teams and management

## Projects

**ITutor App for Til-Qazyna** | *Docker Swarm, FastAPI, vLLM, Unsloth* — ITutor

- Configured and deployed **LLM, ASR, and TTS APIs** on **3 servers with an A10 GPU each**, and set up **monitoring and load balancing systems** to serve **100 users**
- Optimized **ASR, LLM, and TTS model inference**, cutting **median latency** by **2x for ASR**, **3x for LLM**, and **4x for TTS** with **10 concurrent users**
- Fine-tuned an **LLM for the Kazakh language** using LoRA/QLoRA, improving model quality by **10%** on benchmarks

**Datalake AI Platform for Samruk-Energy** | *LangGraph, LangChain, Langfuse, Docling*

- Fully **self-hosted AI platform** with a **ChatGPT-style interface**, automating **6 business use cases**, including:

- **Generative BI**: An agent that **generates SQL queries** in **under one minute**, enabling non-technical users to quickly access and analyze corporate data through a natural-language chat interface
- **Financial Reporting**: An agent that **generates complex Excel** reports from **corporate databases** in **under one minute** based on user requests, enabling rapid exploration of financial forecasting scenarios
- **Tender Analysis**: An agent that analyzes uploaded tender documents against **27 compliance checks** and generates reports highlighting identified discrepancies and violations

**Master's Thesis** | *NumPy, Pandas, Matplotlib, PyTorch, PyGame* GitHub

- Published and won **III place** (certificate) at **conference «2025 IEEE AITU: Digital Generation»**
- Built an **evolutionary reinforcement learning** model for decentralized multi-agent systems, where agents controlled by a **Neural Network** evolved through **genetic algorithms** to optimize survival in a 2D grid environment

## Skills

**Programming**: Python, SQL, Bash, HTML/CSS/JavaScript
**Machine Learning & LLM**: NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, Hugging Face Transformers, Unsloth
**Agentic AI & RAG**: LangGraph, LangChain, Pydantic, MCP, OpenAI API, Anthropic API, Qdrant, Docling, ASR (faster-whisper), Orpheus TTS
**Data & Analytics**: MS SQL Server, Databricks, PySpark, Power BI (Power Query, DAX), advanced Excel, VBA, Azure
**Infra & Serving**: FastAPI, vLLM, LiteLLM, Docker, Docker Swarm, Kubernetes (kubectl), Traefik
**MLOps & Observability**: MLflow, W&B, Prometheus, Grafana, Locust, Langfuse, LangSmith, LLM-as-judge
**Developer Tools**: Git, CI/CD (GitLab CI), Claude Code, Codex, context engineering, custom skills/rules
**Languages**: English (B2), Russian (Native)